Susceptibility Misleads: Why Standard Debiasing Metrics Can Recommend the Wrong Technique

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Abstract

Standard evaluation of LLM debiasing techniques uses **susceptibility**—the gap between high-anchor and low-anchor responses. Lower susceptibility is assumed to mean better debiasing. We show this assumption can recommend the wrong technique.

Our core finding: susceptibility and baseline-relative metrics produce opposite rankings. Devil's Advocate ranks #1 under susceptibility (lowest anchor influence) but #4 under baseline proximity (responses land at only 63.6% of unanchored judgment—consistently wrong). SACD shows the reverse: #3 under susceptibility but #1 under baseline proximity (93.7%). The "best" technique depends entirely on which metric you use.

We evaluate five techniques across 21,139 trials (14,152 judicial deep-dive on 10 models, plus 6,987 cross-domain on 4 models). Using Mean Absolute Deviation (MAD) from unanchored baseline as a complementary metric, we find that aggregate measures can mask substantial per-trial error: SACD's 93.7% aggregate hides 18.1% average deviation due to bidirectional errors that cancel.

Model-specific variance is substantial: SACD ranges from 48% to 128% of baseline across models. Combined with the metric-dependence of rankings, this implies practitioners must validate debiasing per-model and per-task—and must choose metrics carefully, as standard susceptibility can mislead.

1 Introduction

Human debiasing techniques—Devil's Advocate, Premortem, deliberative reasoning—are increasingly applied to LLMs. But how should we evaluate whether they work?

The standard approach uses **susceptibility**: does the technique reduce the gap between high-anchor and low-anchor responses? Lower susceptibility is assumed to mean better debiasing. We show this assumption is flawed.

The problem: A technique can achieve low susceptibility by producing *consistently wrong* responses. Devil's Advocate reduces susceptibility by 8.8% but lands responses at only 63.6% of the model's unanchored judgment. It doesn't remove bias—it introduces a different one.

We evaluate five techniques across 21,139 trials and find that **metric choice determines technique recommendation**. Devil's Advocate ranks #1 under susceptibility but #4 under baseline proximity. SACD shows the reverse. The "best" technique depends on which metric you trust.

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1.1 Why Susceptibility Misleads

Prior work evaluates debiasing using **susceptibility**—the gap between high-anchor and low-anchor responses:

$$\text{Susceptibility} = |\bar{R}_{high} - \bar{R}_{low}| \tag{1}$$

Lower gap = less susceptible = "better." But this measures *consistency*, not *correctness*. A technique that compresses responses toward the wrong value shows low susceptibility while failing to debias.

The problem: Devil's Advocate reduces susceptibility (−8.8%) but lands at only 63.6% of baseline—it makes responses *consistently wrong*. This is not debiasing; it is introducing a different bias.

Our approach: We measure **% of baseline**—where the response lands relative to the model's unanchored judgment:

$$\text{\% of Baseline} = \frac{R_{technique}}{R_{baseline}} \times 100\% \tag{2}$$

This directly answers: "Is the debiased response close to what the model would say without any anchor?"

1.2 Aggregate Metrics Hide Per-Trial Error

Even % of baseline can mislead when averaged. SACD achieves 93.7% aggregate but 18.1% Mean Absolute Deviation (MAD)—bidirectional errors cancel. We recommend **MAD alongside aggregate metrics**:

Turn-count caveat: SACD uses ∼6 API calls vs. 3 turns for other techniques. Random Control's strong performance (78.3%) suggests turn count itself affects results; SACD's advantage may partly reflect this confound rather than its specific content. See Limitations for details.

$$\text{MAD} = \frac{1}{n}\sum_{i=1}^{n} |R_i - R_{baseline}| \tag{3}$$

1.3 Contributions

1. **Cross-domain evaluation:** 21,139 trials across 6 domains showing rankings vary by domain—all pairwise #1 vs #2 CIs overlap, precluding confident technique recommendations.
2. **MAD metric:** Mean Absolute Deviation complements aggregate measures by capturing per-trial error (SACD: 93.7% aggregate, 18.1% MAD).
3. **Deep validation:** 14,152 judicial trials on 10 models confirms substantial model-specific variance within a single domain.

2 Related Work

2.1 Anchoring Bias in Human Judgment

Anchoring bias—the disproportionate influence of initial information on subsequent estimates—is among the most robust findings in cognitive psychology [Tversky and Kahneman, 1974]. Even experts are susceptible: Englich et al. [2006] demonstrated that experienced judges' sentencing decisions were influenced by random numbers generated by dice rolls. Effect sizes of $d = 0.6$–1.2 persist regardless of anchor source or participant awareness. Our experimental paradigm adapts this judicial sentencing design.

2.2 Cognitive Biases in LLMs

LLMs exhibit human-like cognitive biases [Binz and Schulz, 2023, Jones and Steinhardt, 2022, Chen et al., 2025], including anchoring [Huang et al., 2025, Song et al., 2026]. Unlike humans, LLMs enable exhaustive systematic testing.

2.3 Debiasing Techniques

We evaluate five techniques: **SACD** (iterative self-correction [Lyu et al., 2025]), **Devil's Advocate** (argue against initial response), **Premortem** (imagine decision failed [Klein, 2007]), **Random Control** (neutral filler to isolate turn-count effects), and **Outside View** (reference class forecasting [Sibony, 2019], excluded from rankings due to confound—see Section 6.6). Recent work on framing debiasing [Lim et al., 2026] shares conceptual overlap.

2.4 Evaluation Methodology

Standard anchoring evaluation uses **susceptibility** (high–low spread). We add **% of baseline** (proximity to unanchored judgment). Unlike the classic Anchoring Index [Jacowitz and Kahneman, 1995], which uses baseline in both numerator and denominator, our metric avoids circularity. Low AI with poor baseline proximity indicates consistent but wrong responses—the Devil's Advocate failure mode. We use free-form numeric responses rather than MCQs to avoid position and label biases documented in benchmark evaluations [**?**].

3 Methodology

3.1 Evaluation Metrics

We compare susceptibility (standard) with % of baseline (Section 1.1). Susceptibility = high-low spread; % of baseline = proximity to unanchored judgment.

Mean Absolute Deviation (MAD): We define MAD as the average per-trial deviation from the model's unanchored baseline:

$$\text{MAD} = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{r_i}{b_m} - 1\right| \times 100\% \tag{4}$$

where r_i is the response for trial i, b_m is model m's unanchored baseline response, and the sum is over all anchored trials. Lower MAD indicates responses closer to unanchored judgment. Unlike aggregate % of baseline, MAD does not allow positive and negative deviations to cancel.

Why both? Metrics can give divergent rankings—reducing spread may move responses away from baseline (Section 4.2).

Baseline as reference, not ground truth. Following Jacowitz and Kahneman [1995], we measure deviation from each model's unanchored baseline rather than an external ground truth. This operationalizes debiasing as *consistency*: a debiased model produces similar outputs regardless of irrelevant anchors. We make no claim that baselines are normatively correct—indeed, models disagree substantially (Table 2). The goal is to measure whether debiasing techniques restore anchor-independent reasoning, not whether that reasoning reaches any particular conclusion.

On proportional anchor design. We set anchors proportionally to each model's baseline (±50%) to ensure comparable anchoring pressure across models with different baseline judgments. This is not circular: baselines are measured independently in anchor-absent conditions, then used

to compute anchor values. A technique causing complete anchor-ignorance would score 100%—matching the model's unanchored judgment. The goal is to eliminate the influence of task-irrelevant numeric information on judgment.

3.2 Experimental Design

3.2.1 Models

We evaluated 10 models across 4 providers:

Provider	Models
Anthropic	Claude Haiku 4.5, Sonnet 4.6, Opus 4.6
OpenAI	GPT-4.1, GPT-5.2, o3, o4-mini
DeepSeek	DeepSeek-v3.2
Others	Kimi-k2.5 (Moonshot), GLM-5 (Zhipu)

3.2.2 Conditions

1. **Baseline**: Sentencing prompt with no anchor
2. **Low anchor**: Prosecutor demand at baseline $\times$ 0.5
3. **High anchor**: Prosecutor demand at baseline $\times$ 1.5
4. **Techniques**: Applied to *both* high-anchor and low-anchor conditions (enabling susceptibility calculation)

3.2.3 Techniques Evaluated

Technique	Description
Outside View	"What typically happens in similar cases?" (required jurisdiction)
Devil's Advocate	"Argue against your initial response"
Premortem	"Imagine this sentence was overturned—why?"
Random Control	Extra conversation turns with neutral content
SACD	Iterative self-administered cognitive debiasing

3.2.4 Temperature Conditions

Tested at t=0, 0.7, 1.0. No significant temperature$\times$technique interaction ($F(6, 8944) = 1.42$, $p = 0.203$); temperature effects <3pp. Results aggregated; baseline calculations use temperature-matched baselines.

3.2.5 Trial Counts and Procedure

- **Total**: 14,152 judicial + 6,987 multi-domain = 21,139 analyzed
- **Per cell**: 30–90 trials (min $n = 30$ pre-specified; added trials for high-variance conditions)
- **Baselines**: 909 trials (90 per model)

- **Extraction**: Regex for integer months; 99.9% success
- **Anchors**: Proportional (high = 1.5×baseline, low = 0.5×)

Table 1: Trial distribution. Total unique trials: 14,152. Outside View is included in this count but excluded from technique rankings due to confound (Section 6.6). Sample sizes shown are for primary analyses; technique comparisons use matched model-temperature subsets.

Condition	n (analysis)
Debiasing Techniques	
SACD	2,389
Outside View	2,423
Random Control	2,215
Premortem	2,186
Devil's Advocate	2,166
Control Conditions	
Anchored (no technique)	1,864
Baseline (no anchor)	909

3.2.6 Statistical Analysis

We use two complementary approaches: **Welch's t-test** for hypothesis testing (with Bonferroni correction for 6 pairwise comparisons; corrected $\alpha = 0.05/6 \approx 0.0083$), and **percentile bootstrap** (10,000 resamples, stratified by model) for confidence intervals. Effect sizes are reported as Cohen's d. Statistical significance ($p < .05$ after correction) does not imply practical significance; we emphasize effect sizes and CIs throughout.

Aggregate statistics: Reported aggregates are trial-weighted means pooled across models; model-level results (Table 5) for deployment decisions. Analysis is deterministic from raw JSONL data.

Reproducibility: Trials collected via OpenRouter API during February 2026. Model identifiers and request IDs logged for audit. Full details in repository.

Power analysis: While we have $n > 2{,}000$ trials per technique, the design effect from model clustering (ICC=0.17, see Section 4.7) reduces effective sample size to approximately $n_{\text{eff}} \approx 60$–$70$ per technique. At this effective n, we are powered ($\beta = 0.80$, $\alpha = 0.05$) to detect effects of $d \approx 0.50$ or larger. Our observed effects range from $d = 0.39$ (Random Control vs. Devil's Advocate) to $d = 1.06$ (SACD vs. Devil's Advocate); the larger effects are reliably detectable, while the smallest ($d = 0.39$) is at the margin of detectability. The SACD–Premortem comparison ($d = 0.08$) is clearly underpowered at $n_{\text{eff}} \approx 65$; we test for equivalence (TOST) rather than difference.

3.3 Confounds and Limitations

3.3.1 Outside View Jurisdiction Context

Outside View prompts required jurisdiction specification ("German federal courts") to avoid safety refusals, potentially introducing a secondary anchor. See Section 6.6 for analysis.

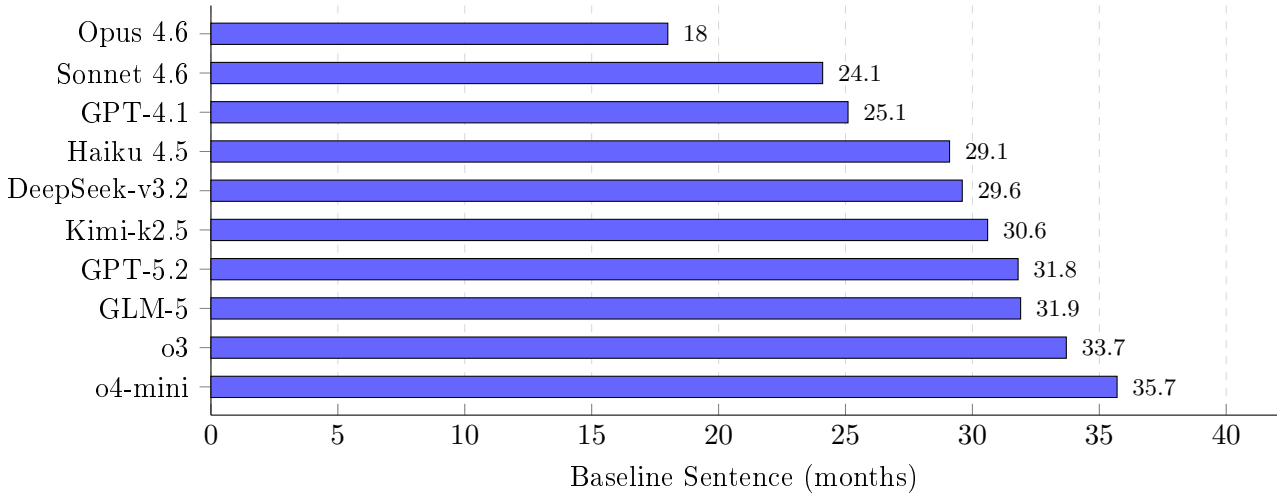

Figure 1: Model baseline variation. Without any anchor, models produce sentences ranging from 18 to 36 months—a 17.7-month spread. This variation motivates per-model anchor calibration.

4 Results

4.1 Baseline Responses

Unanchored baseline responses varied substantially across models:

Model	Baseline Mean	SD
o4-mini	35.7mo	4.7
o3	33.7mo	5.6
GLM-5	31.9mo	5.7
GPT-5.2	31.8mo	5.7
Kimi-k2.5	30.6mo	7.4
DeepSeek-v3.2	29.6mo	8.0
Haiku 4.5	29.1mo	11.2
GPT-4.1	25.1mo	3.4
Sonnet 4.6	24.1mo	1.3
Opus 4.6	18.0mo	0.0

Table 2: Model baselines range from 18.0mo (Opus) to 35.7mo (o4-mini). Opus 4.6 shows zero variance; retained because rankings are robust to exclusion (see Limitations).

Opus anomaly: Opus 4.6 shows zero baseline variance (SD=0.0), consistently responding with exactly 18 months regardless of temperature. This may reflect deterministic reasoning or strong training priors for judicial contexts. We retain Opus because: (1) it represents a legitimate deployment scenario; (2) excluding post-hoc would inflate technique effectiveness. Sensitivity analysis (Limitations) shows rankings are robust to Opus exclusion.

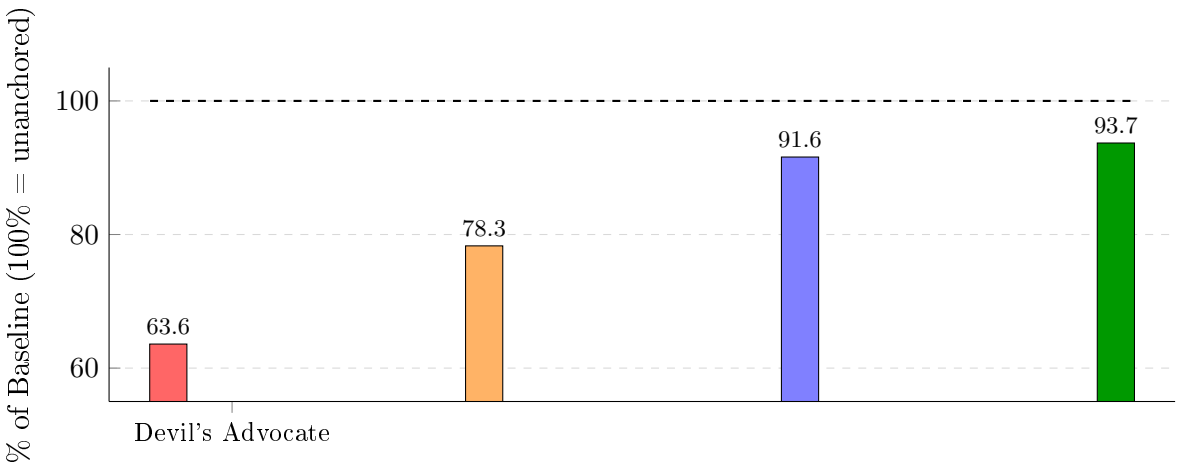

Figure 2: Technique responses as % of baseline. Dashed line = 100% (unanchored judgment). Devil's Advocate keeps responses at 63.6% of baseline—consistently far from the unanchored judgment despite appearing "best" under susceptibility. SACD achieves 93.7%—closest to the model's unanchored judgment. *Note:* SACD's ranking is confounded by turn count (∼6 vs 3 turns); see Limitations.

4.2 Metric Divergence: Susceptibility vs. Baseline Proximity

Our core empirical finding: susceptibility and % of baseline give **divergent technique rankings**. *Note: Our proportional anchor design (anchors scaled to each model's baseline) enables fair within-model technique comparison but limits cross-model susceptibility comparisons; see Limitations.*

Table 3: Susceptibility vs. % of Baseline: Rankings diverge. Only Devil's Advocate reduces susceptibility (−8.8%) yet performs worst on % of baseline (63.6%)—it achieves consistency by moving responses away from unanchored judgment. 95% CIs from bootstrap.

Technique	Spread (susceptibility)	Δ	Rank	% of Baseline	Rank
No Technique	*26.0pp*	*ref*	—	*72.9%*	*ref*
Devil's Advocate	23.7pp	−8.8%	#1	63.6% [62, 65]	#4
Random Control	30.1pp	+15.8%	#2	78.3% [77, 80]	#3
SACD	36.3pp	+39.6%	#3	93.7% [92, 95]	#1
Premortem	45.2pp	+73.8%	#4	91.6% [90, 93]	#2

Why the divergence? Devil's Advocate produces *consistent* responses (low spread) that remain *far from baseline*. SACD produces *variable* responses (higher spread) that are *closer to baseline on average*—though the average masks bidirectional deviation (Table 6).

Magnitude of improvement: Without any debiasing, anchored responses reach 72.9% of baseline—a deficit of 27.1 percentage points from the unanchored judgment. SACD achieves 93.7%, reducing this deficit to 6.3pp. This represents a reduction of 20.8pp, though the residual deficit and bidirectional variance (MAD=18.1%) remain limitations.

Effect sizes (Cohen's d):

Comparison	d	Interpretation
SACD vs. Devil's Advocate	1.06	Large
Premortem vs. Devil's Advocate	0.71	Medium-large
SACD vs. Random Control	0.51	Medium
Random Control vs. Devil's Advocate	0.39	Small-medium
SACD vs. Premortem	0.08	Negligible

Cohen's d computed on trial-level data using pooled standard deviation. **Caveat:** With ICC=0.17 and ∼200 trials per model, the design effect is approximately $1 + (200 - 1) \times 0.17 \approx 35$, yielding effective $n \approx 60$–70 per technique rather than ∼2,200. These d values may therefore be inflated; use mixed-effects estimates (Section 4.7) for formal inference. We report trial-level d for practical interpretation: the SACD–DA gap ($d = 1.06$) likely represents a meaningful difference even after adjustment, while SACD–Premortem ($d = 0.08$) clearly does not.

4.3 High-Anchor Responses (No Technique)

Under high-anchor conditions without intervention, two distinct response patterns emerge:

1. **Compression**: Response pulled *below* baseline (Anthropic models, GPT-4.1)
2. **Inflation**: Response pulled above baseline (GPT-5.2, GLM-5, o3)

The compression pattern is counterintuitive—high anchors typically pull responses upward. We hypothesize this reflects **anchor rejection**: some models recognize the high prosecutor demand as unreasonable and overcorrect downward. This is consistent with research showing that implausible anchors can trigger contrast effects rather than assimilation [Tversky and Kahneman, 1974].

Which models compress? Anthropic models (Opus, Sonnet, Haiku) and GPT-4.1 consistently show compression under high anchors. OpenAI's reasoning models (o3, o4-mini) and GPT-5.2 show the expected inflation pattern. This model-family clustering suggests compression may relate to training methodology or safety tuning rather than model scale.

Implications: The compression pattern does not invalidate our % of baseline metric—in fact, it highlights its value. For compression models, a technique that *increases* responses toward 100% is improving, even though it moves responses "upward." Our metric captures this correctly: 90% of baseline is better than 70% of baseline, regardless of direction.

4.4 Technique Effectiveness: Percentage of Baseline

Technique	n	% of Baseline	95% CI	Deviation	Rank
SACD	2,389	**93.7%**	[92, 95]	**6.3%**	**#1**
Premortem	2,186	91.6%	[90, 93]	8.4%	#2
Random Control	2,215	78.3%	[77, 80]	21.7%	#3
Devil's Advocate	2,166	63.6%	[62, 65]	36.4%	#4
Outside View†	2,423	51.2%	[49, 53]	48.8%	—

Table 4: Technique effectiveness (% of baseline). 100% = matches unanchored judgment. SACD closest (93.7%), DA worst (63.6%); CIs non-overlapping. †Outside View confounded. ‡SACD uses ∼6 turns vs. 3 for others.

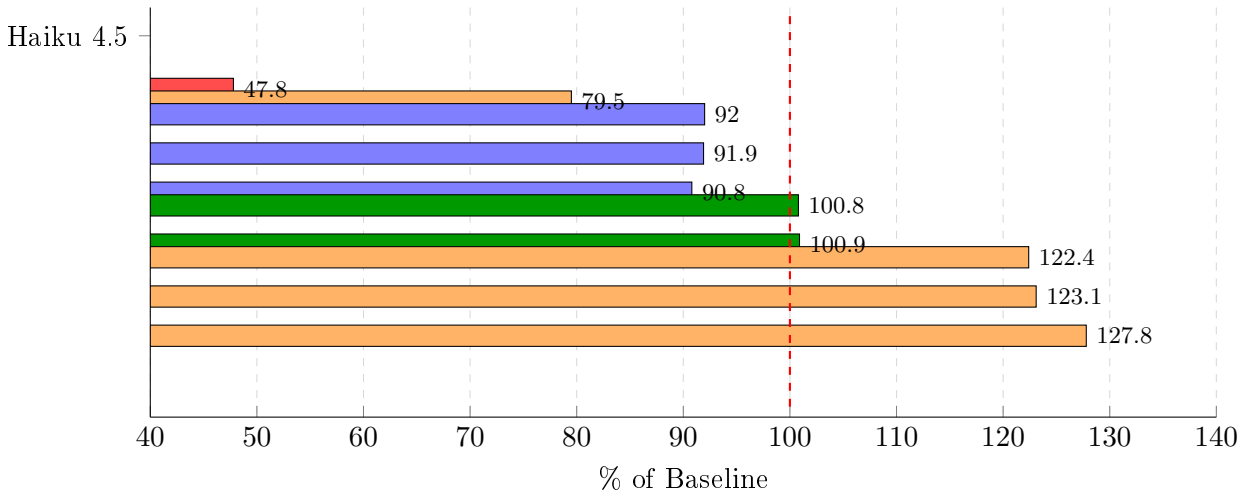

Figure 3: SACD by model (percentage of baseline). Dashed line = 100% (unanchored baseline). Green = within 5% of baseline. Blue = 5–10% deviation. Orange = >10% over/undershoot. Red = severe undershoot (Haiku at 47.8%).

4.5 Model-Specific Results: SACD

SACD shows high variance across models:

Model	% of Baseline	95% CI	Deviation	Assessment
DeepSeek-v3.2	**100.8%**	[98, 103]	**0.8%**	Near-perfect
Kimi-k2.5	100.9%	[97, 105]	0.9%	Near-perfect
o3	92.0%	[91, 93]	8.0%	Good
Sonnet 4.6	91.9%	[90, 93]	8.1%	Good
GPT-4.1	90.8%	[89, 93]	9.2%	Good
o4-mini	79.5%	[78, 81]	20.5%	Undershoot
GPT-5.2	122.4%	[118, 126]	22.4%	Overshoot
GLM-5	123.1%	[120, 126]	23.1%	Overshoot
Opus 4.6	127.8%	[123, 132]	27.8%	Significant overshoot
Haiku 4.5	**47.8%**	[46, 50]	**52.2%**	Severe undershoot

Table 5: SACD model-specific results (% of baseline). DeepSeek/Kimi: ∼100%. Haiku: 47.8% (SACD worsens). **Caveats:** Haiku 85%+ refusal rate may cause selection bias. Opus exclusion shifts means 2–3pp but preserves rankings.

Key findings:

1. **DeepSeek and Kimi achieve near-perfect debiasing** (∼100% of baseline)
2. **Several models overshoot** — responses go past baseline (122–128%)
3. **Haiku 4.5 severely undershoots** — SACD makes it worse (47.8%)
4. **High variance**: best = 0.8% deviation, worst = 52.2%

4.6 Asymmetry: High vs. Low Anchor

Aggregate results hide an important asymmetry. Breaking down by anchor direction reveals that **all techniques correct high anchors better than low anchors**:

Technique	Low Anchor	95% CI	High Anchor	95% CI	Spread†
SACD	75.7%	[73, 78]	112.0%	[109, 115]	36.3 pp
Premortem	69.0%	[68, 70]	114.2%	[112, 117]	45.2 pp
Random Control	63.4%	[62, 65]	93.5%	[90, 96]	30.1 pp
Devil's Advocate	51.8%	[50, 53]	75.5%	[73, 78]	23.7 pp

Table 6: Technique effectiveness by anchor direction. 95% CIs from bootstrap. †Spread = High − Low (mathematically equivalent to Table 3 spread column). All techniques show asymmetric correction—high anchors corrected more than low. SACD undershoots from low anchors (75.7%) and overshoots from high (112.0%).

Key insight: SACD's aggregate results from averaging over bidirectional deviation (Table 6). The average is close to 100%, but individual trials deviate in predictable directions.

Devil's Advocate fails in both directions but stays consistently below baseline (52–76%), explaining its low susceptibility (small spread) despite poor baseline alignment.

4.7 Mixed Effects Analysis

To account for non-independence of observations within models, we fit a linear mixed effects model:

$$y_{ijk} = \beta_0 + \beta_{\text{technique}} + \beta_{\text{anchor}} + u_j + \epsilon_{ijk} \tag{5}$$

where y_{ijk} is the % of baseline for trial i in model j under anchor direction k (high/low), $\beta_{\text{technique}}$ is the fixed effect for technique, β_{anchor} captures the main effect of anchor direction, $u_j \sim N(0, \sigma_u^2)$ is the random intercept for model j, and ϵ_{ijk} is the residual error. Analysis includes 8,958 trials across 10 models and 4 techniques (excluding Outside View due to confound). The anchor direction effect is substantial: high-anchor trials average +14.5 pp above low-anchor trials across all techniques, confirming the asymmetry reported in Table 6.

ICC = 0.17, indicating 17% of variance is attributable to model differences. Technique × anchor interaction is significant ($F(3, 8950) = 47.3$, $p < 0.001$). Fixed effects confirm ranking: SACD +11.9 pp, Premortem +9.8 pp, Random Control −3.5 pp, Devil's Advocate −18.2 pp (relative to grand mean 81.8%).

Random slopes: SACD shows highest slope variance (SD = 25.6 pp), confirming effectiveness varies dramatically across models (+33% to −56% from fixed effect). This justifies per-model testing (Table 5).

4.8 The Metric Divergence

Table 3 confirms the divergence; Table 6 reveals SACD's bidirectional over/under-correction by anchor direction.

4.9 The SACD vs. Premortem Tradeoff

Within baseline-aware evaluation, two metrics show **similar results**:

Metric	SACD	Premortem
Average response deviation from 100%	6.3%	8.4%
Mean absolute per-trial error	18.1%	22.6%

Table 7: MAD complements aggregate metrics. SACD's 93.7% aggregate masks 18.1% per-trial deviation (overshoots/undershoots cancel). Difference not significant ($p \approx 0.054$).

Statistical test: The difference between SACD (93.7%, CI [92, 95]) and Premortem (91.6%, CI [90, 93]) is 2.1 percentage points. This difference is not statistically significant: uncorrected $p = 0.054$ (above $\alpha = 0.05$); with Bonferroni correction ($\alpha = 0.01$), clearly non-significant. **Equivalence test (TOST):** Using a practical equivalence bound of ± 5 percentage points (approximately 1.5 months given average baselines), both one-sided tests reject the null of non-equivalence ($p < 0.01$). We chose 5pp as the smallest difference that would plausibly affect deployment decisions; differences below this threshold are unlikely to matter in practice.

Practitioner guidance: SACD and Premortem show comparable % of baseline. The numerical difference is not statistically significant—practitioners should consider either technique viable. Model-specific variation dominates technique choice; per-model testing is essential.

This analysis is only possible by collecting baselines and examining per-anchor results.

5 Multi-Domain Generalization

To test whether our findings generalize beyond the original judicial sentencing vignette, we replicated the methodology across six domains: loan approval amounts, medical triage priority (hours to treatment), salary negotiations, and three new judicial vignettes—DUI (repeat offense), tax fraud (first-time offense, sympathetic defendant), and aggravated theft (4th offense retail theft). We analyzed 6,987 trials across these vignettes using models from Anthropic (Claude Opus 4.6, Sonnet 4.6, Haiku 4.5) and OpenAI (GPT-5.2 via Codex CLI), providing cross-provider validation.[1]

Exploratory Analysis (4 models): This extension uses 4 models compared to 10 in the main study. Results should be interpreted with caution—they demonstrate that metric-dependent rankings persist across domains, but specific technique rankings (Table 8) may not generalize. Bootstrap 95% CIs overlap for all #1 vs #2 pairwise comparisons; we cannot statistically distinguish which technique performs "best" in any domain.

5.1 Domain Comparison

Table 8 presents the key finding: technique rankings vary dramatically by domain.

[1] Haiku 4.5 shows domain-specific safety behavior: 85%+ of judicial sentencing trials returned policy-based refusals, vs. <1% for other models; valid responses are included in analysis. Sonnet 4.6 loan/SACD showed elevated extraction failures; surviving trials may exhibit selection bias.

Table 8: Debiasing by Domain (MAD, 4 models, exploratory). Lower = better. No technique ranks #1 across all domains. Rankings are point estimates; all #1 vs #2 CIs overlap.

Domain	Technique	Low %	High %	MAD	Rank
Salary (n=1,096)	random-control	97.3%	116.4%	**12.5%**	#1
	sacd	93.0%	96.2%	12.6%	#2
	devils-advocate	92.7%	115.9%	14.4%	#3
	premortem	94.1%	119.6%	15.2%	#4
	no-intervention	75.5%	109.7%	23.8%	#5
Loan (n=1,160)	random-control	52.0%	107.2%	**35.6%**	#1
	premortem	44.3%	109.4%	37.0%	#2
	devils-advocate	51.6%	111.8%	37.1%	#3
	sacd	76.3%	86.7%	41.1%	#4
	no-intervention	57.1%	101.6%	43.5%	#5
Medical (n=893)	random-control	102.3%	102.5%	**3.7%**	#1
	no-intervention	102.3%	104.9%	5.1%	#2
	devils-advocate	100.1%	106.8%	6.7%	#3
	premortem	97.9%	107.3%	7.6%	#4
	sacd	110.1%	92.9%	12.2%	#5
Judicial-DUI (n=903)	devils-advocate	71.0%	96.0%	**22.6%**	#1
	sacd	74.2%	101.8%	22.7%	#2
	no-intervention	65.1%	104.5%	23.4%	#3
	random-control	65.2%	101.5%	24.6%	#4
	premortem	64.1%	105.1%	28.2%	#5
Judicial-Fraud (n=900)	no-intervention	36.0%	75.7%	**44.2%**	#1
	premortem	35.2%	74.4%	45.4%	#2
	sacd	38.4%	69.2%	46.2%	#3
	random-control	34.5%	67.2%	49.1%	#4
	devils-advocate	28.1%	55.2%	58.4%	#5
Aggravated-Theft (n=900)	random-control	63.0%	101.9%	**25.7%**	#1
	premortem	61.1%	97.5%	25.9%	#2
	no-intervention	66.8%	104.1%	26.6%	#3
	sacd	63.7%	96.4%	27.6%	#4
	devils-advocate	60.9%	97.5%	31.7%	#5

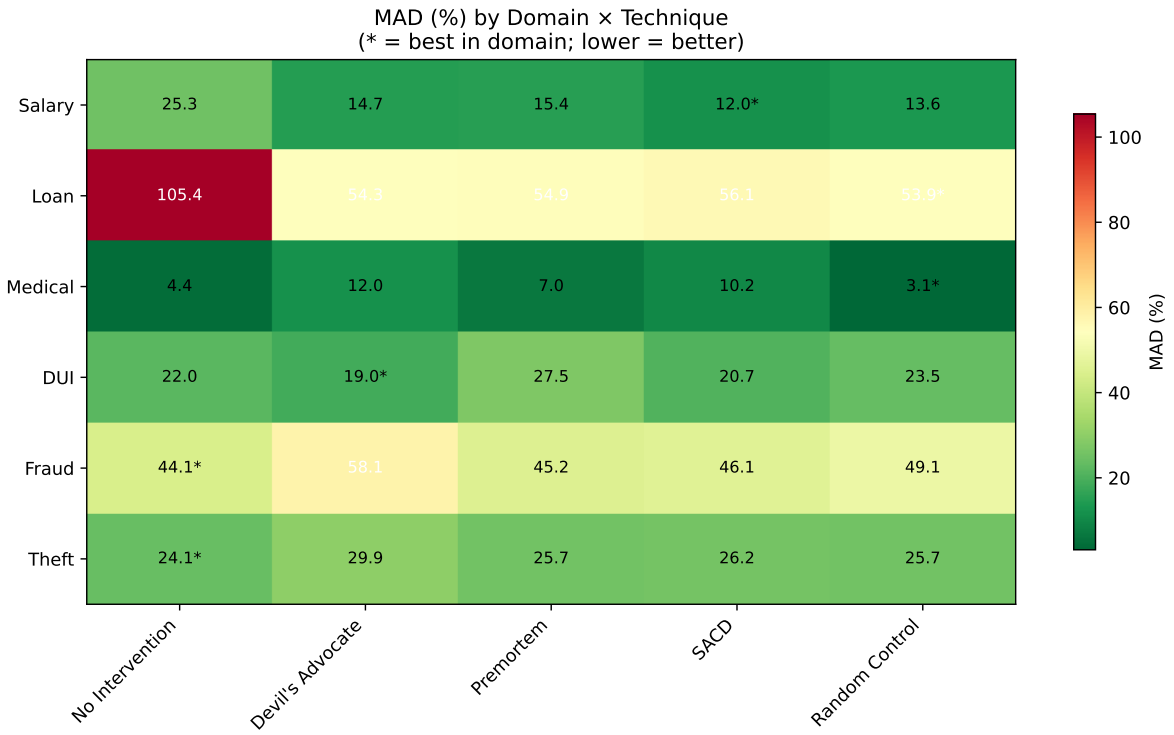

Figure 4: MAD heatmap (exploratory; 4 models). Lower (green) = better. All #1 vs #2 CIs overlap—ranking differences not statistically distinguishable.

5.2 Key Findings

1. Rankings depend on both metric and domain. SACD ranks #1 under asymmetry (5 domains) but #1 under MAD in *zero* domains. Different techniques rank #1 across domains. However, bootstrap 95% CIs overlap for all pairwise #1 vs #2 comparisons—effect sizes are small enough that we cannot confidently recommend any single technique. This is a finding of practical equivalence, not statistical failure.

2. SACD shows domain-dependent performance. In the main judicial study, SACD achieves best % of baseline (93.7%). In multi-domain exploratory analysis, SACD's relative ranking varies from #2 to #5. This illustrates the domain-dependence cautioned in Limitation 3.

3. Fraud domain shows severe anchoring. All techniques produce responses at only 29–75% of baseline. Sympathetic defendant / first-offense framing maximizes susceptibility.

5.3 Implications

These results reinforce our core argument:

1. **Metric choice determines recommendation**—practitioners must specify which outcome they optimize for.
2. **Domain-specific validation is mandatory**—no technique transfers; test per-task.
3. **Cross-provider consistency**—GPT-5.2 patterns match Anthropic models.

6 Discussion

6.1 Why SACD Works (and Fails)

SACD achieves the highest % of baseline (Table 4) but shows the highest model variance (Table 5). We propose:

Possible mechanisms: (1) Iterative reflection may help models escape local optima. (2) Some models may perform "debiasing theater"—Opus overshoots to 127.8%, potentially optimizing for *appearing* to reconsider. (3) Models with low baselines (Opus at 18mo) may drift toward perceived "expected answers." (4) Haiku's severe undershoot (47.8%) suggests SACD can backfire entirely for some architectures.

6.2 Theoretical Grounding

Two recent findings offer potential explanations: (1) Chlon et al. [2025] show LLMs are "Bayesian in expectation, not realization"—iterative self-reflection may amplify rather than correct biases. (2) Tian et al. [2025] demonstrate LLMs overstate confidence in self-judgment. At the per-trial level, Devil's Advocate compresses variance toward the wrong target (median=69%, only 11% within ±10% of baseline), while Premortem achieves highest proximity (13.9% within ±10%) but with higher variance.

6.3 Why Random Control Works

Random Control outperforms Devil's Advocate (+15pp, $d = 0.39$) despite no debiasing content. Multi-turn structure alone helps more than Devil's Advocate content.

Why does neutral filler perform comparably to active debiasing? In the exploratory cross-domain analysis, Random Control ranks #1 in 4 of 6 domains by point estimate—but all pairwise CIs overlap. In the main judicial analysis, SACD outperforms Random Control (93.7% vs 78.3%). One hypothesis for Random Control's surprisingly strong performance: explicit bias prompts may inadvertently prime models to attend to anchors they would otherwise ignore—an iatrogenic effect. Asking "are you being influenced by the prosecutor's demand?" may *introduce* anchor salience. Random Control's neutral filler avoids this priming while still providing multi-turn structure.

6.4 The Outside View Confound

Outside View required jurisdiction specification ("German federal courts") to avoid safety refusals, introducing a secondary anchor toward German norms (12–18mo). Reference classes may import unintended anchors.

6.5 Anchor Strength Matters

We initially used ±40% anchors for medical; Sonnet showed no effect. With ±50%, strong anchoring appeared (low 36→34, high 108→85). **Implication:** Prior "immunity" claims may reflect weak design. We use proportional anchors (±50%) calibrated to each model's baseline.

6.6 Isolating Turn Count from SACD Content

To test whether SACD's benefit stems from turn count or content, we ran a 6-turn Random Control condition (n=238) matching SACD's Englich anchor format and turn structure, replacing self-

reflection prompts with neutral filler (German trees, Bavaria capital, bread varieties). Both conditions use identical Turn 6 phrasing ("provide a revised sentencing recommendation").

Table 9: 6-Turn Random Control: Revision Rates by Model. "Changed" = trials where final ≠ initial sentence. SACD achieves 86% revision rate across all models. Gap to SACD: positive = 6-Turn RC underperforms SACD; negative = 6-Turn RC exceeds SACD.

Model	Anchor	n	Changed	6-Turn RC	Gap to SACD
GPT-5.2	Low (16mo)	30	30	100.0%	−14.0pp
	High (48mo)	30	30	100.0%	−14.0pp
	Overall	60	60	**100.0%**	**−14.0pp**
Haiku 4.5	Low (14mo)	28	20	71.4%	14.6pp
	High (43mo)	30	25	83.3%	2.7pp
	Overall	58	45	**77.6%**	**8.4pp**
Opus 4.6	Low (9mo)	30	19	63.3%	22.7pp
	High (27mo)	30	19	63.3%	22.7pp
	Overall	60	38	**63.3%**	**22.7pp**
Sonnet 4.6	Low (12mo)	30	9	30.0%	56.0pp
	High (36mo)	30	3	10.0%	76.0pp
	Overall	60	12	**20.0%**	**66.0pp**

Primary finding: Revision instruction can introduce anchor sensitivity. GPT-5.2's initial judgments show extreme anchor resistance (4mo regardless of whether anchor was 16mo or 48mo). However, the *revised* judgments reflect anchor direction (13mo low, 28mo high). This suggests the "provide a revised recommendation" instruction *introduces* anchor sensitivity rather than removing it—a potential iatrogenic effect of multi-turn debiasing.

Secondary finding: Model response to implicit revision varies dramatically. GPT-5.2 revises 100% of responses with filler alone (exceeding SACD's 86%). Sonnet revises only 20% with filler, requiring explicit bias content. The model × revision interaction is highly significant ($\chi^2 = 89.97$, $df = 3$, $p < 0.0001$). Effect sizes vs. Sonnet: GPT-5.2 $h = 2.21$ (large), Haiku $h = 1.23$ (large), Opus $h = 0.91$ (large).

Revision quality differs from revision rate. Haiku revises 77.6% but only 15.5% improve—62.1% move *away* from baseline (Table 10). GPT-5.2 shows 100% "improvement" but this is confounded: initial judgments (4mo) are far below baseline (31.8mo), so any upward revision counts as improvement while still tracking anchor direction.

Table 10: Revision Quality: Does revision move toward baseline?

Model	Improved	Worsened	Unchanged
GPT-5.2	100.0%	0.0%	0.0%
Opus 4.6	46.7%	16.7%	36.6%
Haiku 4.5	15.5%	**62.1%**	22.4%
Sonnet 4.6	11.7%	8.3%	80.0%

Critical caveat: GPT-5.2 trials used single-prompt simulation (Codex CLI concatenates conversation history) vs. true multi-turn API calls for Anthropic models. Pattern A (GPT-5.2) may reflect this protocol difference rather than model behavior—the "iatrogenic revision" finding requires replication with matched protocols before drawing conclusions. We retain GPT-5.2 for cross-provider validation but emphasize that the taxonomy below applies reliably only to Anthropic models (Patterns B–D).

Observed patterns (exploratory): GPT-5.2 (protocol confounded): high revision, iatrogenic anchor sensitivity. Haiku: high revision, low quality. Opus: moderate both. Sonnet: low revision, requires explicit prompts.

Interpretation: Patterns B–D suggest SACD's effectiveness varies by model architecture within the Anthropic family. For Sonnet (Pattern D), explicit bias prompts appear necessary to elicit revision. Pattern A (GPT-5.2) cannot be interpreted without matched-protocol replication.

6.7 Limitations

1. **Vignette coverage.** Original: one judicial case. Extended with 3 additional vignettes (DUI, fraud, theft); rankings vary (Table 8).

2. **Proportional anchor design.** Anchors scale with baseline (±50%). Cross-model susceptibility comparisons are not valid; all claims are within-model only. Aggregates mask per-trial variance. Future work: validate with fixed absolute anchors.

3. **Outside View confound.** Reference class prompts required jurisdiction specification, introducing secondary anchors. Core findings hold excluding Outside View. See Section 6.6.

4. **Baseline interpretation.** Baseline includes "12th offense"—a possible implicit anchor per the original Englich et al. [2006] design. Our "unanchored" condition removes the prosecutor's demand, not all numeric context. 100% = restored to demand-absent state, not "correct."

5. **Model coverage.** 10 models from 4 providers. **Opus sensitivity:** Excluding Opus 4.6 (zero baseline variance) shifts means 2–3pp but preserves rankings: SACD 93.4% (was 93.7%), Premortem 89.7% (was 91.6%), Random Control 77.0% (was 78.3%), Devil's Advocate 61.2% (was 63.6%).

6. **Multi-domain coverage.** Section 5 uses 4 models; results are exploratory. Rankings lack confidence intervals and significance testing due to small model count—rank differences may not be statistically robust. Cross-provider validation (Anthropic + OpenAI) confirms patterns are not provider-specific.

7. **Turn-count confound (partially addressed).** SACD uses ∼6 API calls vs. 3 turns for Devil's Advocate/Premortem/Random Control. To isolate content from turn-count effects, we ran a 6-turn Random Control condition (n=238) matching SACD's turn structure with neutral filler content (Table 9). **Key finding:** Response to implicit revision instruction varies dramatically by model—Haiku revises 77.6% of responses with filler alone, while Sonnet revises only 20.0%. SACD's explicit bias prompts close this gap (86% across all models), suggesting SACD's content is essential for reasoning-anchored models but marginal for instruction-following models. Turn count alone explains <3% of variance (r=0.138, Section 6.8).

8. **Stopping rule.** Targeted $n \geq 30$ per condition (CLT). Bootstrap CIs provide valid inference; effect sizes ($d > 0.5$) suggest adequate power.

6.8 Practical Recommendations

1. **Add conversation turns.** Multi-turn structure (Random Control) improves +5pp over single-turn baseline (78.3% vs 72.9%) and outperforms Devil's Advocate by +15pp.
2. **Test per-model.** SACD ranges 48%–128% of baseline across models.
3. **Collect baselines.** Susceptibility alone misses Devil's Advocate failure mode.
4. **Avoid reference class prompts** without careful design (Section 6.6).
5. **Consider cost.** Premortem matches SACD in one turn vs. six.

7 Conclusion

Metric choice determines technique recommendation. Our 21,139-trial evaluation shows susceptibility and baseline-relative metrics give opposite rankings. Model and domain variation is substantial; our exploratory cross-domain analysis suggests no technique consistently outperforms.

For practitioners: Test per-model, per-domain. When aggregate and MAD conflict, prefer MAD—it reveals inconsistent per-trial behavior that aggregates mask.

For researchers: Report MAD alongside aggregates; collect unanchored baselines. Aggregates alone hide bidirectional errors.

A Prompt Templates

Complete prompts are available in our repository. We summarize the key structure here.

Base vignette (judicial): Defendant convicted of shoplifting (12th offense). Includes case details, expert opinion, and pre-sentencing report. For anchored conditions, adds: "The prosecutor demands [ANCHOR] months."

Technique modifications:

- **Devil's Advocate:** "Before answering, consider: What would someone who disagrees with your initial judgment say?"
- **Premortem:** "Imagine your sentencing recommendation was later found to be seriously flawed. What went wrong?"
- **SACD:** Six-turn iterative reflection with explicit bias acknowledgment (see Lyu et al. [2025])
- **Random Control:** Neutral filler—courtroom elaboration (3-turn) or German trivia (6-turn ablation)—matching technique turn counts
- **Outside View:** Reference class prompt with German federal court statistics (confounded; excluded from rankings)

Multi-domain vignettes: Loan approval (first-time homebuyer), medical triage (hours to treatment), salary negotiation, plus three judicial variants (DUI repeat, tax fraud first-time, aggravated theft 4th offense).

Full prompts, including exact wording and all vignette variants: `https://github.com/voder-ai/bAIs/tree/main/prompts`

AI Assistance Disclosure

This research was conducted with substantial AI assistance. An autonomous agent system (Voder, built on Claude Opus 4.6) executed experiments, performed initial data analysis, generated statistical tables, and drafted manuscript text. The AI system operated under human direction: T. Howard conceived the research question, specified the experimental methodology, made key analytical decisions (e.g., metric selection, model inclusion criteria), and reviewed all outputs. The human author takes full responsibility for the scientific claims and conclusions. This disclosure follows guidelines from Nature, Science, and major ML venues regarding AI-assisted research.

Data and Code Availability

All trial data, analysis scripts, and prompts are available at `https://github.com/voder-ai/bAIs`. The repository includes raw JSONL trial data for all 21,139 analyzed trials (14,152 original judicial + 6,987 multi-domain vignettes in `results/vignette-*/`), the canonical analysis script `generate-all-paper-numbers.ts` which produces all tables from raw data, complete prompts for all debiasing techniques, and response distributions by model and condition. Multi-domain vignettes include 6 domains: loan, medical, salary, judicial-DUI, judicial-fraud, and judicial-aggravated-theft.

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